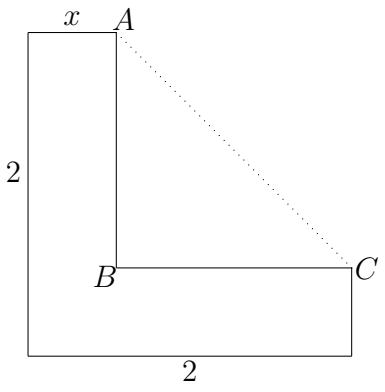
NMC 2010 A-LEVEL PAPER 2 QUESTIONS

1. If $x^2 + \frac{1}{x^2} = 3$, find the value of $\frac{(x+1)^2}{x^2}$.
2. Compute the exact value of the infinite series,

$$-\frac{4}{3} + \frac{1}{3^2} - \frac{4}{3^3} + \frac{1}{3^4} - \frac{4}{3^5} + \dots$$

3. When 60 minutes elapse on a correct clock, 62 minutes register on clock F (fast) and only 56 minutes register on clock S (slow). If later in the day clock F reads $8:00PM$ and clock S reads $7:00PM$ what was the correct time when the two clocks were originally set?
4. If $x - 2$ divides the polynomial $x^4 - 3ax^2 + (5a - 2)x - 16$ without leaving a remainder, find the value of a.
5. Consider the function $f(x)$ which is such that $f(mn) = f(m - n)$ for all real numbers m and n, if $f(4) = 2$, find the value of $f(-3) + f(6)$.
6. If $x > 0$, and $\log_a b = 4$, find the value of $\frac{\log_a x}{\log_{ab} x}$?
7. How many distinct pairs of integers (x, y) satisfy the function $y = \frac{x+12}{2x-1}$?
8. Suppose that $a < b$ and that $3^2 + 4^2 + 5^2 + 12^2 = a^2 + b^2$ is satisfied by only one pair of positive numbers (a, b), What is the sum of the numbers a and b?
9. A six sided die has faces labeled 1 through 6. It is weighted so that a three is three times as likely to be rolled as a one, a three and a six are equally likely, and a one, a two, a four and a five are equally likely. What is the probability of rolling a three?
10. In the figure shown below, if the area of the letter "L" equals the area of the triangle ABC. What is the length x of the ends of the letter L?



SECTION B : 10 **marks each**

11. (a) If $g\left(\sqrt{\frac{1+x}{1-x}}\right) = 5x$, Determine the value of $g(2)$.

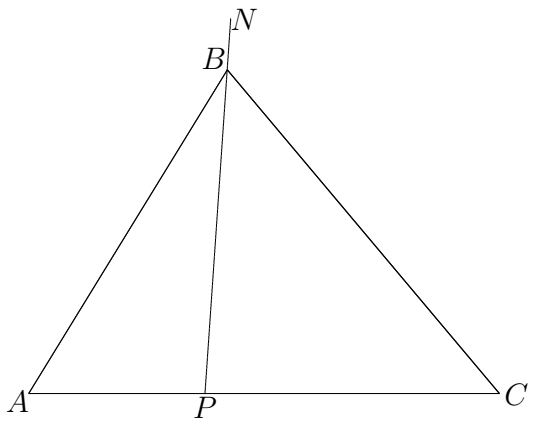
    (b) The number $\sqrt{19+\sqrt{336}}$ can be written in the form $\sqrt{a}+\sqrt{b}$ where a and b are whole numbers and $a > b$. What is the value of $a - b$?

12. A swimming team consists of eight members only, two of which are boys. The coach wants to take a delegation from the team to a special swimming camp. If the delegation must have either five or six members and must include at least one boy. How many ways are there to select the delegation?

13. Bill and Jill are hired to paint a line on a road. If Bill works by himself, he could paint the line in B hours. If Jill works by herself, she could paint the line in J hours. Bill starts painting the line from one end and one hour later Jill begins from the other end. They both work until the line is painted. Determine the number of hours that Bill works in terms of B and J.

14. According to a free fall model an object released from rest will fall $16t^2$ feet, t seconds after it's dropped. Suppose that two marbles are dropped one after the other from a cliff that is 64 feet high and that the second marble is released the instant the first marble has fallen exactly one foot. According to the free fall model, how far above the ground is the second marble at the instant the first strikes the ground.

15. The triangle ABC has sides $BC = 13, AC = 14$ and $AB = 15$ as shown below. Line N bisects angle B and crosses side AC at P. Find the distance from A to P.



NMC 2010 A-LEVEL PAPER 2 SOLUTIONS

SECTION A

1. Let $y = x^2$. Then, $x^2 + \frac{1}{x^2} = 3 \Rightarrow y + \frac{1}{y} = 3 \Rightarrow y^2 - 3y + 1 = 0 \Rightarrow y = \frac{3 \pm \sqrt{5}}{2}$.

$$\begin{aligned}
\frac{(x+1)^2}{x^2} &= \left(1 + \frac{1}{x}\right)^2 \\
&= 1 + \frac{1}{x^2} + \frac{2}{x} \\
&= 1 + \frac{2}{3 \pm \sqrt{5}} \pm \frac{2\sqrt{2}}{\sqrt{3 \pm \sqrt{5}}} \\
&= \frac{3 \pm \sqrt{5} + 2 \pm 2\sqrt{2} \times \sqrt{3 \pm \sqrt{5}}}{3 \pm \sqrt{5}} \\
&= \frac{5 \pm \sqrt{5} \pm 2\sqrt{6 \pm 2\sqrt{5}}}{3 \pm \sqrt{5}}.
\end{aligned}$$

Note that in all there are four solutions.

2.

$$\begin{aligned}
-\frac{4}{3} + \frac{1}{3^2} - \frac{4}{3^3} + \frac{1}{3^4} - \frac{4}{3^5} + \ldots &= -4\left[\frac{1}{3} + \frac{1}{3^3} + \frac{1}{3^5} + \ldots\right] + \left[\frac{1}{3^2} + \frac{1}{3^4} + \ldots\right] \\
&= -4\left[\frac{\frac{1}{3}}{1 - \frac{1}{9}}\right] + \left[\frac{\frac{1}{3^2}}{1 - \frac{1}{3^2}}\right] = -\frac{3}{2} + \frac{1}{8} = -\frac{11}{8}.
\end{aligned}$$

3.

Fast Clock	$8:00PM$	$6:58PM$	$5:56PM$	$4:54PM$	$3:52PM$	$2:50PM$
Slow Clock	$7:00PM$	$6:04PM$	$5:08PM$	$4:12PM$	$3:16PM$	$2:20PM$

Fast Clock	$1:48PM$	$12:46PM$	$11:44AM$	$10:42AM$	$9:40AM$
Slow Clock	$1:24PM$	$12:28PM$	$11:32AM$	$10:36AM$	$9:40AM$

The correct time on both clocks was set at $9:40AM$.

4. $x-2$ divides $x^4-3ax^2+(5a-2)x-16$ implies that $x=2$ solves the equation $x^4-3ax^2+(5a-2)x-16=0$. That is, $16-12a+2(5a-2)-16=0 \Rightarrow a=-2$.

5. Since $f(4)=2$, we have $f(0)=f(2-2)=f(2\times 2)=2$ by taking $m=n=2$. Likewise,

$$\begin{aligned} f(-3) &= f(1\times -3)=f(1--3)=f(4)=2. \\ f(0) &= f(3\times 0)=f(3-0)=f(3)=2. \\ f(0) &= f(1\times 0)=f(1-0)=f(1)=2. \\ f(6) &= f(3\times 2)=f(3-2)=f(1)=2. \end{aligned}$$

   Thus, $f(-3)+f(6)=2+2=4$.

6. Let us show first of all that $\log_{ab} x=\frac{\log_a x}{\log_a(ab)}$. Let $y=\log_{ab} x \Rightarrow (ab)^y=x$. Taking logs to base a gives $y\log_a(ab)=\log_a x$ from which we deduce that $y(\log_{ab} x)=\frac{\log_a x}{\log_a(ab)}$. Therefore,

$$\frac{\log_a x}{\log_{ab} x} = \frac{\log_a x}{\frac{\log_a x}{\log_a(ab)}}=\log_a(ab)=\log_a a+\log_a b=1+4=5.$$

7. Only pairs $(1,13),(13,1),(0,-12),(-12,0)$ and $(-2,-2)$ do satisfy the relation $y=x+12x-1$.

8. You can see that $3^2+4^2+5^2+12^2=5^2+13^2$. Therefore $a=5$ and $b=13$.

9. Let $P(k)$ be the probability of rolling side $k, k=1,2,\ldots,6$ and let $P(1)=p$. Since $\sum_{k=1}^{6}=1$ we have that $p+p+3p+p+p+3p=1 \Rightarrow p=\frac{1}{10}$. Therefore, $P(3)=3p=\frac{3}{10}$.

10.

$$\text{Area of triangle} = \frac{1}{2}\cdot b\cdot h=\frac{1}{2}(2-x)(2-x)=\frac{1}{2}(2-x)^2.$$

$$\text{Area of Letter } L = 2x+x(2-x).$$

   Since the two areas are equal, we have

$$2x+(2-x)=\frac{1}{2}(2-x)^2 \Rightarrow 3x^2-12x+4=0 \Rightarrow x=\frac{6\pm 2\sqrt{6}}{3}.$$

SECTION B : 10 **marks each**

11. (a) When $2=\sqrt{\frac{1+x}{1-x}} \Leftrightarrow x=\frac{3}{5}$. Therefore, $g(2)=5\times\frac{3}{5}=3$.

(b) Let $\sqrt{a}+\sqrt{b}=\sqrt{19+\sqrt{336}}$. We want to find a and b.

$$(\sqrt{a}+\sqrt{b})^2=(\sqrt{19+\sqrt{336}})^2$$

$$\begin{aligned}
&\Leftrightarrow\ a+b+2\sqrt{ab}=19+\sqrt{336}\\
&\Leftrightarrow\ a+b+2\sqrt{ab}=19+\sqrt{336}\\
&\Leftrightarrow\ a+b=19 \text{ and } \sqrt{ab}=\sqrt{84}
\end{aligned}$$

By letting $b=\frac{84}{7}$, we get a quadratic equation $a+\frac{84}{a}=19$ whose solution is 12 or 7. Taking $a=12$ gives $b=7$ and taking $a=7$ gives $b=12$. The value of $a-b$ is either 5 or -5.

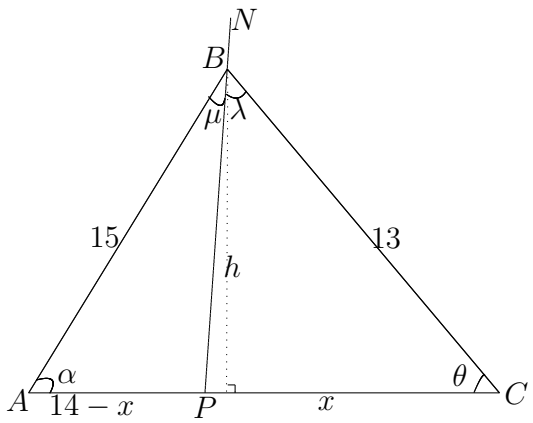
12. If the delegation consists of 5 members, then, there are $\begin{pmatrix}2\\1\end{pmatrix}\begin{pmatrix}6\\4\end{pmatrix}+\begin{pmatrix}2\\2\end{pmatrix}\begin{pmatrix}6\\4\end{pmatrix}=50$ ways to select them.

If the delegation consists of 6 members, then, there are $\begin{pmatrix}2\\1\end{pmatrix}\begin{pmatrix}6\\5\end{pmatrix}+\begin{pmatrix}2\\2\end{pmatrix}\begin{pmatrix}6\\4\end{pmatrix}=27$ ways to select them. There are $50+27$ ways of selecting a delegation of either 5 or 6.

13. Let the distance of the line on the road be D. Jill's speed of work is $\frac{D}{J}$ per hour and Bill's speed is$\frac{D}{B}$. per hour. The combined rate of work for both $\frac{D}{B}+\frac{J}{B}$ per hour. Since Jill starts to paint one hour later, the section of the line painted by both Bill and Jill is $D-\frac{D}{B}$. Let t be the time it takes to paint this remaining portion. Then $\left(\frac{D}{B}+\frac{D}{J}\right)t=D-\frac{D}{B}\ \Rightarrow\ t=\frac{(B-1)J}{B+J}$. In total, Bill paints for $t+1=\frac{(B-1)J}{B+J}+1=\frac{B(J+1)}{B+J}$ hours.

14. The first marble falls to the ground after time $t=\sqrt{\frac{64}{16}}=2$ seconds. The second marble is dropped when the first one has fallen one foot. Time required to fall one foot is $\sqrt{\frac{1}{16}}=\frac{1}{4}$. So, by the time the first marble fall, the second will have traveled for $2-\frac{1}{4}=\frac{7}{4}$ seconds and it will have fallen a height h given by $h=16t^2=16\times\frac{49}{64}=49$. The distance left to hit the ground is therefore $64-49=15$ feet.

15.



Find h using triangles BMC and ABM.

$$h^2 = 13^2 - x^2 \tag{18}$$
$$h^2 = 15^2 - (14 - x)^2 \tag{19}$$

Solving (18) and (19) gives $x = 5$ and $h = 12$.

$$\angle BCM = \sin^{-1}\left(\frac{12}{13}\right) = 67.3^0$$

$$\angle BAM = \sin^{-1}\left(\frac{12}{15}\right) = 53.1^0$$

$$\angle ABC = 180 - (67.3 + 53.1) = 59.6^0$$

$$\angle ABP = \angle PBC = \frac{1}{2}\angle ABC = 29.8^0.$$

$$\angle CPB = 180 - (29.8 + 67.3) = 82.9^0.$$

Distance $PM = \frac{12}{\tan 82.9} = 1.495$ and distance $AP = 14 - (5 + 1.495) = 7.505$.